# Muhammad Umer

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## PROFESSIONAL SUMMARY

Senior Full Stack Software Developer with 7+ years of experience working with teams from 6 different countries and delivering high-performance, cloud-native applications across healthcare, fintech, Real Estate and AI sectors. Expert in architecting scalable microservices using **Node.js/NestJS/Express, Java/Spring Boot, Python/Flask** and building dynamic frontends with **React/Next.js** and **React Native**. Proven record in leading technical teams, integrating advanced AI services (LLMs, Computer Vision), and designing **offline-first** systems with **real-time data** synchronization. Accomplished in deploying production-grade solutions on **GCP and AWS**, ensuring security, high availability, and regulatory compliance.

## CORE COMPETENCIES

Full-Stack Development | Frontend & Application Performance | Technical Leadership & Mentoring | Cloud Infrastructure & DevOps | Restful & Event-Driven Architecture | Offline-First & Real-Time Systems | Microservices & API Design | Application Security & Secure Architecture | AI/ML Service Integration | Enterprise Solution Architecture

## SKILLS

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|---|---|
| **Cloud & Infrastructure** | Google Cloud Platform (GCP - GKE, Cloud Run, Cloud Functions, Cloud Storage, Firestore, Cloud SQL, Pub/Sub, VPC, IAM, Logging, Monitoring), AWS Lambda, Azure, Terraform, Docker, Kubernetes, Vercel |
| **Backend Architecture** | Node.js, NestJS, Express.js, Fastify, Flask, Microservices, Serverless Architecture |
| **DevOps & Automation** | Terraform, Infrastructure as Code (IaC), GitHub Actions, GitLab CI, CI/CD Automation, Jenkins, Jest, Playwright, Postman |
| **Integration Services** | Pub/Sub, Event-Driven Architecture, Twilio, RabbitMQ, Kafka, System Integration, API Architecture |
| **Security & Auth** | IAM, OAuth 2.0, JWT, MFA, RBAC, Rate Limiting, Session Management, OWASP, System Security, Security Baseline Design |
| **Databases & Data Flow** | Firebase Firestore, Firebase Real-time Database, MongoDB, PostgreSQL, ElasticSearch, MySQL, SQL Server, Redis, GraphQL |
| **Governance & Planning** | Technical Leadership, Architecture Decisions, High Availability/Disaster Recovery (HA/DR), Solution Architecture, Jira, Confluence |
| **Languages** | JavaScript, TypeScript, Python, C#, Java, HTML5/CSS |
| **Frontend Architecture** | React.js, React Native, Next.js, Flutter, Redux, Zustand, Context API, Tailwind CSS, Material UI, Ant Design, Vite, Webpack |
| **AI & Specialized Architecture** | LLM Integration (llama.cpp, OpenAI), Facial Recognition (Binnah AI), Blockchain (Solana, Ethereum), MetaMask, Vibe Coding (Windsurf, Claude IDE, Claude Code, Agentic AI) |
| **Payment Integration** | Stripe, Poli, Blinq |

## WORK EXPERIENCE

### F Three Technologies

Aug 2023 – Feb 2026

**Lead Software Engineer** — **Sharjah, UAE (Remote)**

- **Cloud Architecture Leadership & Governance:** Directed cross-functional engineering teams of 7 using Agile methodologies; designed end-to-end target cloud solution architectures, landing patterns, and orchestrated zero-downtime server-to-server migrations from legacy systems to distributed GCP microservices architectures.
- **GCP Infrastructure & Automation:** Architected, provisioned, and managed resilient cloud infrastructure on Google Cloud Platform using **Terraform (IaC), Docker, and Google Kubernetes Engine (GKE)** to enforce continuous deployment, high availability, self-healing, and auto-scaling.
- **Event-Driven & Serverless Architecture:** Designed loosely coupled **microservices** and **event-driven** integration pipelines utilizing **GCP Pub/Sub** and cloud messaging patterns to guarantee real-time asynchronous data flow, fault tolerance, and high availability.
- **Cloud Security & IAM Governance:** Architected enterprise identity, security, and baseline compliance mechanisms across GCP environments, incorporating GCP IAM, OAuth 2.0, JWT, MFA, RBAC, and encrypted data transfers aligned with strict security standards.

- **End-to-End System & API Design:** Architected scalable production solution blueprints combining microservices (Node.js, Express.js, NestJS) and cloud APIs for resilient multi-region corporate deployments.
- **AI Service Integration Architecture:** Designed and integrated enterprise AI pipelines into cloud platforms, including real-time biometric vision services (Bio Scan / Binnah AI) and log analytics threat intelligence architectures (Ask Cyfen / ChatGPT + ElasticSearch).
- **CI/CD & Automated Delivery Pipelines:** Architected automated CI/CD delivery pipelines (GitHub Actions, GitLab CI, Jenkins) and automated test execution frameworks (Jest, Playwright) to enforce deployment release governance and operational readiness.
- **Security Audit & Operational Reliability:** Conducted comprehensive security architecture reviews, dynamic analysis (SAST, DAST, API scanning), and edge device security assessments to ensure zero vulnerabilities, OWASP alignment, and GDPR compliance.
- **Architecture Decision Making & Standards:** Established reference architectures, tech evaluation frameworks, and system design patterns to maintain architectural integrity, operational stability, and maintainability across releases.

### F3 Technologies

Oct 2020 – Jul 2023

**Senior Software Engineer** — Pakistan (Onsite)

- **GCP Monolith Legacy Modernization:** Orchestrated end-to-end migration strategies for legacy monolithic applications onto cloud-native **Google Cloud Platform (GCP / GKE)** microservices, integrating federated identity baselines (OAuth 2.0/MFA) and automated CI/CD deployment automation.
- **High Availability & Scalability Architecture:** Designed and implemented distributed GCP containerized architectures (GKE / Compute) engineered to **auto-scale** from 10k requests/sec to over 1M+ requests/sec while meeting strict uptime and reliability metrics.
- **Microservices & Clean Architecture:** Designed resilient full-stack cloud applications and microservice modules (Express.js, NestJS, Next.js); applied **domain-driven** Clean Architecture principles to optimize team efficiency and reduce technical debt.
- **Distributed Data Flow & Persistence Design:** Designed multi-tier, high-availability data flow and backup synchronization architectures across **GCP Firestore, MongoDB, and SQL** databases; integrated secure multi-vendor payment channels **(Stripe, Blinq, Poli)** allowing users to dynamically select their preferred payment method at runtime.
- **Caching & Operational Optimization:** Integrated distributed in-memory caching layers (Redis) within GCP application workloads to reduce database load, minimize latency, and accelerate transaction processing.
- **Frontend & Application Layer Performance:** Trained more than 100 people in **React and React Native**; Engineered modern state management and data-fetching architectures using **React Query**, **SWR**, and **Zustand** with custom hook memoization (**useMemo**, **useCallback**); refactored legacy state workflows to improve client memory performance.
- **Security Baselines & Compliance Hardening:** Enforced enterprise cloud security baselines using **OWASP** standards, secure API designs, and Content Security Policies (CSP); systematically mitigated security risks across **React Native** builds to pass strict audits.

### Freelance

Jan 2020 – Oct 2020

**Full Stack Developer** — Remote

- **Full-Stack & Web3 Architecture:** Developed React.js web applications and integrated decentralized Web3 interfaces with Ethereum/Solana smart contract backends via MetaMask authorization integrations.
- **Smart Contract Solution Design:** Designed and deployed secure blockchain transaction pipelines using Solana Candy Machine programs to ensure high-throughput asset minting and delivery.

### Euroko International

Jun 2019 – Dec 2019

**Software Developer** — Pakistan (Onsite)

- **Application Development & Infrastructure:** Designed and developed a sales management mobile system alongside responsive web applications; directed internal IT infrastructure and database networking support.

## PROJECTS

### Myindici Patient Portal — NZ Top 20 Healthcare

Regulated System Design | View Project

It is serving more than 250 medical practices, Designed and led core feature architectures for a high-concurrency patient portal system serving primary healthcare providers. Built secure modules for insurance workflows, clinical data exchange, and regulatory compliance.

Stack: React.js, React Native, Node.js, Express, Firebase, Firestore, Google Cloud Functions, Cloud Run, PUB/SUB, Redis, PostgreSQL, Terraform, Temporal,Google Cloud Storage, CI/CD, Artifact Registry, Healthcare Compliance

### InnovaCare — Home Care Ecosystem

Serverless Solution Architecture | View Project

Modernized a monolithic platform into a serverless GCP system design with offline-first synchronization capabilities via Firestore. Designed NFC attendance authentication, geofencing capabilities, and a bi-directional SQL-to-NoSQL message bridge utilizing Cloud Pub/Sub.

Stack: Node.js, Express, React-Native, Google Cloud Functions, GKE, Google Cloud Storage, Firestore, VPC, .NET, Pub/Sub, NFC, Artifact Registry

### Vivasta — Community Healthcare Ecosystem

Serverless Solution Architecture | View Project

Modernized a monolithic community healthcare platform into a serverless GCP system design. Engineered cross-platform React Native mobile applications empowering nurses, drivers, and field staff with real-time electronic patient records, referral management pipelines, and localized healthcare logistics tools. Designed offline-first synchronization capabilities via Firestore, geofencing features, and a bi-directional SQL-to-NoSQL message bridge utilizing Cloud Pub/Sub.

Stack: Node.js, Express, React Native, Google Cloud Functions, GKE, Google Cloud Storage, Firestore, VPC, .NET, Pub/Sub, NFC, Geofencing, Artifact Registry

### Zenith AI — Cloud-Native AI Video Content Monetization Platform

Full Stack Solution Architecture | Under Construction

Architected a cloud-native media processing solution that converts prompt inputs into short-form video assets using distributed AI services. Designed automated credit-backed transaction flows, orchestrated multi-stage asynchronous AI media pipelines via Temporal, and delivered real-time progress updates via Firestore state listeners.

Stack: Next.js, Fastify, Temporal, PostgreSQL, Google Kubernetes Engine, Cloud Run, Firestore, Google Cloud Storage, Firebase, Gemini, ElevenLabs, Shotstack, FFmpeg, Terraform, JWT

### InvestLytics — Real Estate Portfolio Management Platform

Fintech/Property Solution Architecture | Under Construction | View Project

Architected and designed a comprehensive WealthTech solution for Portfolio Management System (PMS) providers. Engineered scalable data processing workflows for contract note ingestion, capital gains calculations, and financial analytics (TWRR, Sharpe, VaR). Designed multi-tenant advisor access models, granular RBAC security, and reporting engines.

Stack: Flask, Python, React, MongoDB, Google Cloud Run, Financial Analytics, AI/ML Services

### Grocery — E-Commerce Platform

E-Commerce / Cloud-Native | Under Construction

Developing a production-ready, enterprise-style e-commerce platform to learn and demonstrate modern Java backend development, scalable cloud architecture, secure REST APIs, microservices, and DevOps best practices.

Stack: Java 25, Spring Boot, Spring Security, Spring Data JPA, PostgreSQL, Docker, Kubernetes, Terraform, Google Cloud Platform (GCP), BigQuery, Redis, Kafka, JUnit, Testcontainers

### Ask Cyfen — Kibana + ChatGPT Threat Intelligence

AI Integration Architecture | View Project

Designed an enterprise security integration architecture connecting LLM interfaces with ElasticSearch for natural language log querying inside Kibana. Engineered real-time payload transformation to translate security queries into interactive threat analytics dashboards.

Stack: Node.js, ElasticSearch, Kibana, Plugin, OpenAI API

### Bio Scan — AI Vital Sign Monitor

AI Healthcare Architecture | View Project

Architected a mobile healthcare integration system utilizing Computer Vision APIs (Binnah AI) for contactless facial recognition vital sign monitoring. Deployed scalable micro-container backend services on GCP for global low-latency availability.

Stack: Node.js, React Native, GCP, Cloud Run, Firestore, Binnah AI, Blinq

### Eizhar — Property Management Platform

Enterprise PMS Solution | View Project

Designed an end-to-end property management system architecture supporting automated billing events, real-time communication modules, and asynchronous notification engines powered by an event-driven Pub/Sub design.

Stack: Node.js, Express, React Native, Firestore, Twilio, Pub/Sub, Cloud Functions

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT




**Professional Cloud Architect**
Google Cloud
IN PROGRESS




**Google Cloud Fundamentals: Core Infrastructure**
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**Advanced Android Development Workshop**
COMSATS University Islamabad

## EDUCATION

**B.S. Software Engineering —** COMSATS University Islamabad 2015 – 2019

CGPA: 3.2 / 4.0 · Position holder (2nd & 3rd semester) · Runner-up, Code-A-Than Pre-Coder competition